

# **Week 4 Project Summary — Data Analytics Track**

## ***AnalystLab Africa — HealthConnect Experience Lab***

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### **1. Problem Addressed**

This week, my focus as the Data Analytics track was to develop a solid understanding of the HealthConnect appointment dataset and establish an analytical foundation for investigating why patients miss their scheduled appointments. The goal was to identify which patient, booking, and logistical factors may be associated with no-shows, and to translate these areas into business questions and KPIs that could support the clinic's decision-making later in the project.

### **2. Resources Used**

I worked primarily with HealthConnect\_Appointment\_Data.csv, which contains 5,000 appointment records across 18 fields, alongside HealthConnect\_Data\_Dictionary.csv to understand the meaning of each field and review the stated constraints before analysis.

### **3. Key Observations**

The dataset was structurally sound based on the checks performed. There were no duplicate appointment\_id values, and none of the logical constraints checked during the initial assessment were violated. For example, previous\_no\_shows never exceeded previous\_appointments for any record.

The booking\_lead\_days field was also validated against the difference between booking\_date and appointment\_date, with zero mismatches across all 5,000 records, confirming that the recorded booking lead time is internally consistent with the dates.

Three fields contained missing values. reminder\_channel was missing in 27.3% of records, but further investigation showed that all 1,366 missing values occurred where reminder\_sent = "No". This suggests structural missingness rather than a data-entry error, since no reminder channel exists when no reminder was sent.

The other two fields, distance\_to\_clinic\_km and waiting\_time\_minutes, contained smaller amounts of missing data at 1.8% and 1.2%, respectively. No specific missingness pattern was investigated at this stage, so these fields will be handled cautiously in their respective analyses.

The overall appointment outcome distribution showed a 48.5% No-Show rate, 46.3% Attended rate, and 5.3% Cancelled rate. The high No-Show proportion is a notable characteristic of this synthetic dataset and should not be assumed to represent real-world clinic attendance patterns.

#### **4. Proposed Approach**

Going forward, I plan to calculate and validate the five identified KPIs and examine them against appointment outcomes using cross-tabulations and grouped summary statistics. Any patterns identified will be cross-checked against additional variables where appropriate before conclusions are drawn.

Once the findings have been validated in Python, they will inform the initial Power BI dashboard development in the next phase of the project.

#### **5. Key Considerations**

A few considerations are important as the project develops. The historical fields — `previous_appointments` and `previous_no_shows` — are synthetic and may not represent a complete or coherent real appointment history for each `patient_id`. Therefore, patient-level patterns involving these fields will be interpreted cautiously.

It is also important to distinguish association from causation. The analysis will identify patterns and relationships in the available data, but the dataset does not by itself establish causal reasons for why patients miss appointments.

#### **6. Proposed Focus for Week 5**

Next week, I intend to calculate and validate the five KPIs identified during Week 4, cross-tabulate them against appointment outcomes, investigate the relationships identified in the business questions, and begin initial Power BI dashboard design once the underlying findings have been validated.